

Research question and benchmark

Can sparse regression recover the governing Fokker-Planck equation directly from probability-density data, and remain reliable when the density is estimated from stochastic trajectories?

We use the Ornstein-Uhlenbeck process

$$dX_t = -\kappa X_t dt + \sqrt{2D} dW_t,$$

whose density satisfies the conservative equation

$$\rho_t = \kappa \partial_x(x\rho) + D\rho_{xx}.$$

For computation, we truncate the spatial domain to

$$(x, t) \in [-5, 5] \times [0, 3], \quad X_0 \sim \mathcal{N}(-2, 0.5^2),$$

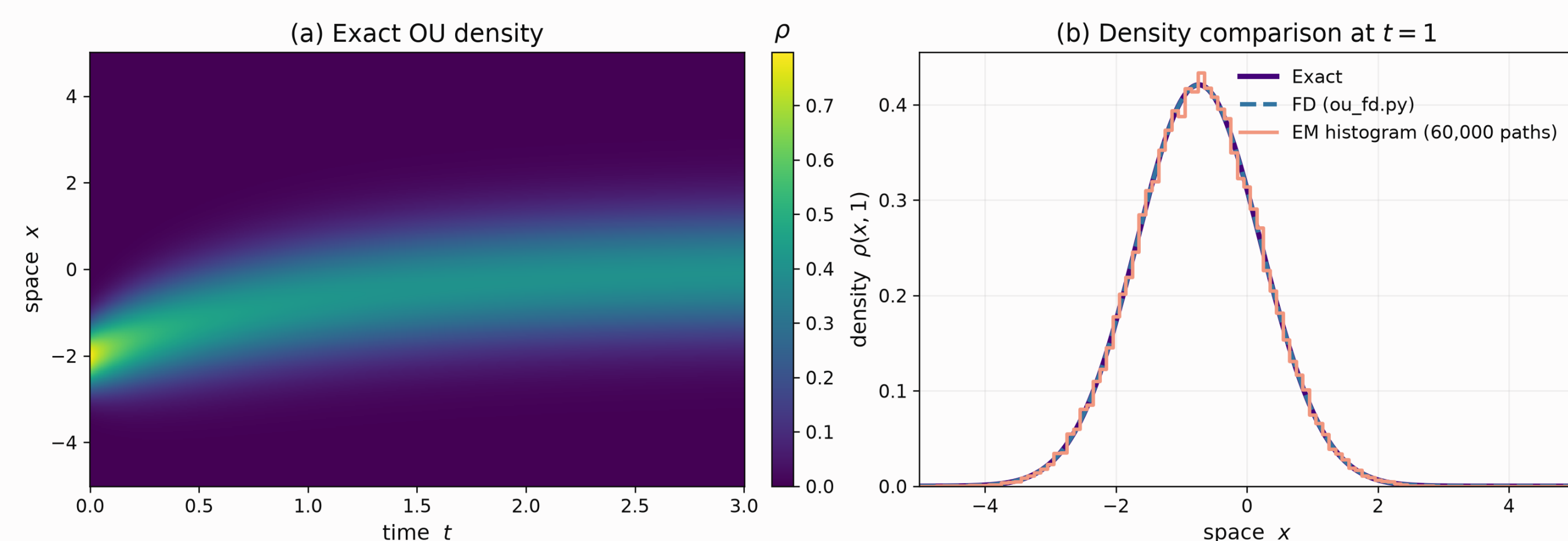
so the initial density data is the associated probability density function.

Here, we set $\kappa = D = 1$, the ground-truth active terms are therefore $\partial_x(x\rho)$ and ρ_{xx} , both with coefficient one.

Density data used

Aspect	Exact	Finite difference	Euler-Maruyama
Construction	analytic OU density	numerical FPE solution	60,000 paths + histogram
Main uncertainty	none(reference)	space-time discretisation	time stepping, sampling and binning

SDE / FPE $\longrightarrow$ density $\rho(x, t)$ $\longrightarrow$ sparse PDE identification



The three data sources agree visually, but the EM histogram is not differentiable pointwise: small bin-to-bin fluctuations are strongly magnified by numerical derivatives.

Why this benchmark is useful

The true equation, exact density and exact time derivative are known. This separates errors caused by **data generation**, **numerical differentiation**, **library design** and **sparse regression** before moving to less controlled data.

Library design and non-uniqueness

The product rule gives an exact dependence between conservative and expanded candidates:

$$\partial_x(x\rho) = \rho + x\rho_x, \quad 0.377\rho + 0.377x\rho_x + 0.623\partial_x(x\rho) = \partial_x(x\rho).$$

Including all three columns makes the library linearly dependent, so individual coefficients are not unique. Prefer a compact conservative library when this structure is known; otherwise interpret results modulo algebraic equivalence.

Sparse identification of the PDE

Ordinary SINDy constructs a candidate matrix and solves

$$\mathbf{b} \approx \Theta \xi, \quad \mathbf{b} = \text{vec}(\rho_t),$$

where ξ is encouraged to be sparse by sequentially thresholded least squares (STLSQ).

A representative strong-form library contains

$$\rho, \rho_x, \partial_x(x\rho), \partial_x(x^2\rho), \rho_{xx}, \rho_{xxx}, \dots$$

Problem: if $\hat{\rho} = \rho + \varepsilon$, finite differences amplify noise approximately as ε/h for first derivatives and ε/h^2 for second derivatives.

Weak SINDy: differentiate the test function, not the data

On each randomly sampled space-time subdomain Ω_k , multiply by a smooth compactly supported test function ϕ_k and integrate by parts:

$$\int_{\Omega_k} \rho_t \phi_k = - \int_{\Omega_k} \rho (\phi_k)_t, \quad \int_{\Omega_k} q_x \phi_k = - \int_{\Omega_k} q (\phi_k)_x,$$

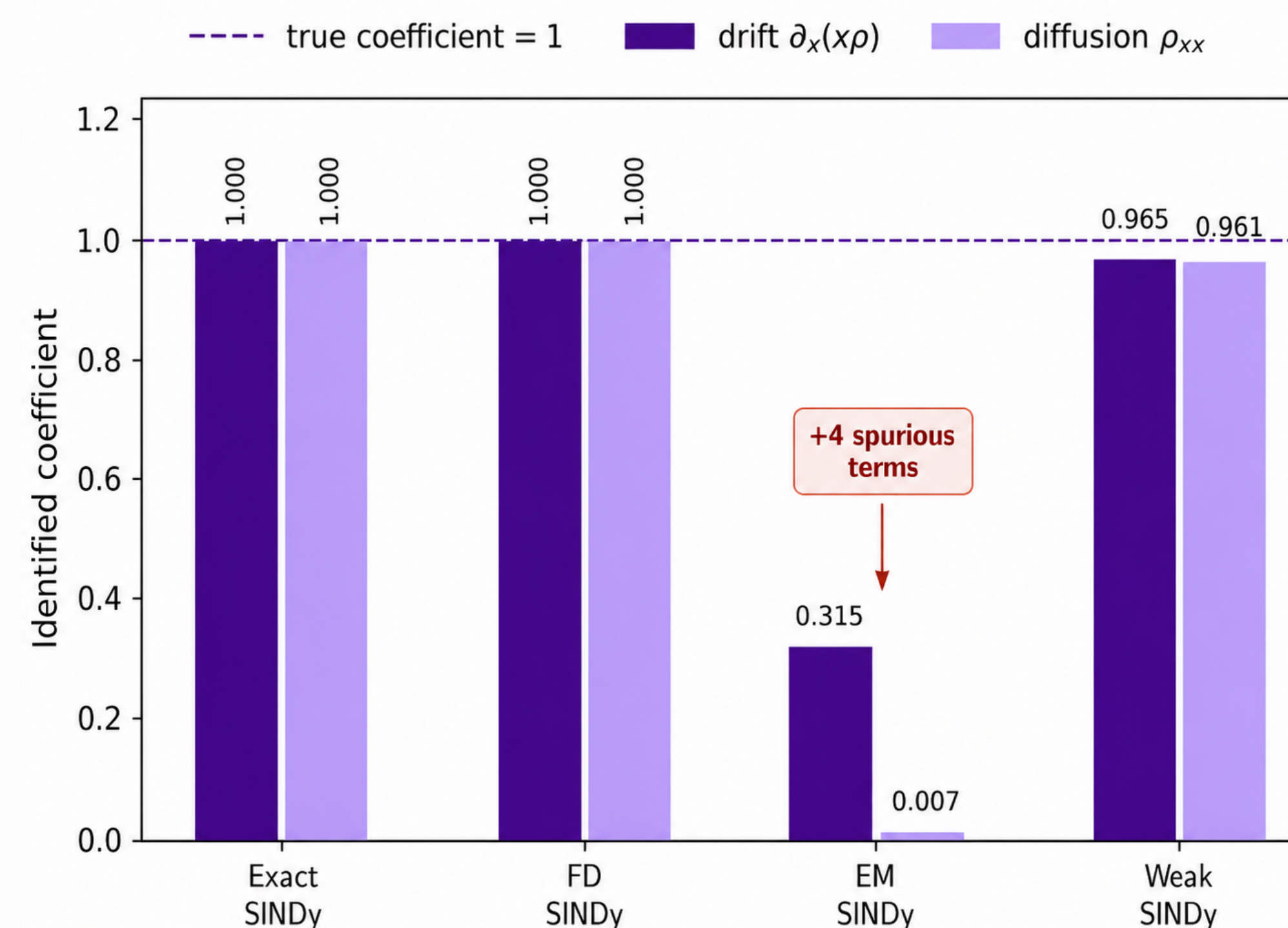
$$\int_{\Omega_k} \rho_{xx} \phi_k = \int_{\Omega_k} \rho (\phi_k)_{xx}.$$

Boundary terms vanish by construction. The observed density appears only inside local weighted integrals, so noise is averaged rather than differentiated.

Lib 1	five selected columns: $\partial_x(x\rho)$, $\partial_x(x^2\rho)$, $\partial_x(x\rho^2)$, $\partial_x(x^2\rho^2)$, ρ_{xx}
Lib 2	all 15 columns obtained from $q \in \{\rho, x\rho, x^2\rho, x\rho^2, x^2\rho^2\}$ with spatial derivative orders 0, 1, 2

Each of the K subdomains supplies one weak regression equation. K controls the number of equations; the subdomain widths control the amount of local averaging.

Core result: weak integration restores coefficient recovery



SINDy + Exact/FD
correct support
coefficients close to 1

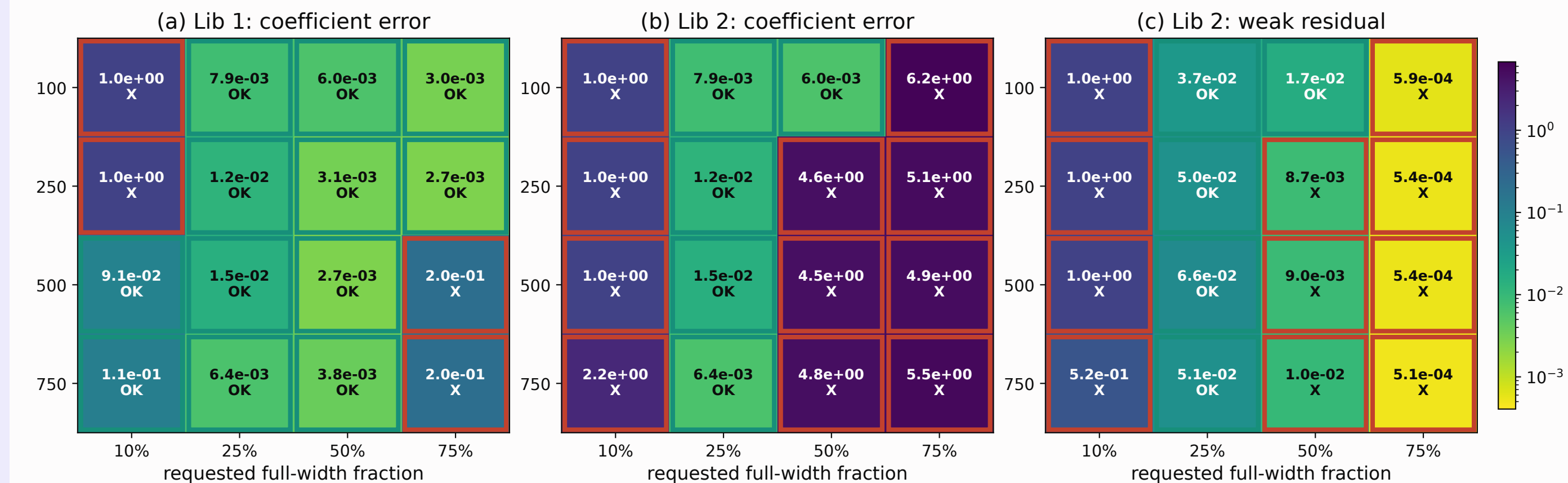
SINDy + EM
diffusion almost lost
+ 4 spurious terms

Weak SINDy + EM
correct support
coefficients close to 1

Weak-SINDy result shown for **Lib 1**, $K = 1000$, 10% requested full widths.

K -subdomain trade-off

Sweep: $K = 100, 250, 500, 750$, with subdomain widths 10%, 25%, 50%, 75% of each axis. Lib 1 and Lib 2 share the same weak target and sampled subdomains for every parameter pair.



- 10% is too small:** local integrals remain noisy. Increasing K helps Lib 1, but cannot reliably rescue Lib 2.
- 25% is the safest common choice:** both libraries recover exactly the two true terms for every K , with coefficient error 0.0064–0.0146.
- 50% favours the reduced library:** Lib 1 remains accurate, whereas Lib 2 retains many spurious terms for $K \geq 250$.
- 75% over-averages:** Lib 2 has coefficient error about 4.9–6.2 even when the weak residual is only about 5×10^{-4} .

Consequently, the smallest residual and the largest K do not necessarily identify the correct PDE.

Conclusions

- SINDy works for smooth densities, but differentiation destabilises EM histogram data.
- Weak SINDy restores the correct OU equation by differentiating test functions instead of data.
- Subdomain size and K must be tuned jointly. In the present sweep, 25% full widths are most robust across both libraries.
- Small residual $\neq$ correct PDE.
- Redundant libraries destroy coefficient identifiability.

References

- [1] Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15):3932–3937, 2016.
- [2] Daniel A. Messenger and David M. Bortz. Weak SINDy for partial differential equations. *Journal of Computational Physics*, 443:110525, 2021.
- [3] Samuel H. Rudy, Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Data-driven discovery of partial differential equations. *Science Advances*, 3(4):e1602614, 2017.